Lenin T

Coimbatore, Tamil Nadu

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Professional Summary

ECE graduate focused on Artificial Intelligence and Machine Learning, with a foundation in Python, data analysis, and machine learning concepts. Familiar with data science libraries and developing skills in deep learning and PyTorch. Hands-on software development experience with FastAPI, REST APIs, and PostgreSQL. Seeking an entry-level AI/ML or GenAI role.

Education

Kumaraguru College of Technology **Nov 2022 – May 2026**

Bachelor of Engineering in Electronics and Communication – CGPA: 7.90 *Coimbatore, Tamil Nadu*

Sri Vivekananda Matriculation Higher Secondary School

Class XII – 86.6% — *Class X – 96.2%* *Coimbatore, Tamil Nadu*

Technical Skills

Programming: Python, SQL, Object-Oriented Programming

Data Science: NumPy, Pandas, Data Preprocessing, Exploratory Data Analysis (EDA), Statistics

Machine Learning: Scikit-learn, Machine Learning Fundamentals, Model Evaluation

Deep Learning: PyTorch (Learning), Deep Learning Fundamentals

Generative AI: LLM Fundamentals, Prompt Engineering

Software and Tools: FastAPI, REST APIs, PostgreSQL, SQLite, Git, GitHub, Docker, Linux, VS Code

Projects

ToLet Coimbatore – Hyperlocal Rental Platform **Apr 2026 – Jun 2026**

Tech Stack: Python, FastAPI, PostgreSQL, React, Cloudinary, Railway

- Built a full-stack rental platform connecting landlords and tenants in Coimbatore through a hyperlocal listing system.
- Developed REST APIs using FastAPI and PostgreSQL for listing CRUD operations, JWT authentication, and photo uploads.
- Implemented a React frontend with map-based listing visualization using Leaflet.js and a mobile-responsive interface.
- Deployed the backend and frontend on Railway with a cloud PostgreSQL database and environment-based configuration.

Vehicle Platooning for Heavy Duty Vehicles **Dec 2025 – Apr 2026**

Tech Stack: ESP32, Arduino UNO, ESP-NOW, RF Communication, Ultrasonic Sensor

- Developed a leader-follower vehicle communication prototype using wireless communication for coordination.
- Integrated ultrasonic obstacle detection and an automatic braking mechanism.
- Implemented distance monitoring to support adaptive spacing between the leader and follower vehicles.
- Developed motor control logic using Arduino UNO and the L298N motor driver for directional movement and speed control.
- Integrated embedded controllers and wireless communication to demonstrate coordinated vehicle movement.

Predictive Maintenance in Factories Using Embedded System **Feb 2025 – Apr 2025**

Tech Stack: ESP32, Ultrasonic Sensor, Gas Sensor, DHT11, Blynk IoT Platform

- Developed an industrial monitoring prototype using ESP32 and environmental sensors for continuous monitoring.
- Implemented gas leakage, temperature variation, and obstacle proximity detection with mobile alerts.
- Integrated sensor-based monitoring and automated safety responses to support early fault detection.

Certifications

- Introduction to Internet of Things – NPTEL (IIT Kharagpur), Elite (91%)
- Fundamentals of Robotics – Zeekers
- Software Defined Vehicles (SDV 101) – Coursera